

Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects*

Liyang Sun[†] and Sarah Abraham[‡]

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Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

Keywords: difference-in-differences, two-way fixed effects, pretrend test

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[†]Department of Economics, MIT, 77 Massachusetts Avenue, Cambridge, MA 02139. Corresponding author; lsun20@mit.edu.

[‡]Cornerstone Research, 699 Boylston St, Boston, MA 02116. The views expressed herein are solely those of the author, who is responsible for the content, and do not necessarily represent the views of Cornerstone Research.

Assumption 3 requires that each cohort experiences the same path of treatment effects. Treatment effects need to be the same across cohorts in every relative period for homogeneity to hold, whereas for heterogeneity to occur, treatment effects only need to differ across cohorts in one relative period. The assumption of treatment effect homogeneity is therefore strong, and in Section 3.4.1, we describe how it can be violated in applied settings.

Our notion of treatment effect homogeneity does not preclude dynamic treatment effects; it only imposes that cohorts share the same path of treatment effects. The related literature sometimes formulates restrictions on the dynamics of treatment effects as another notion of treatment effect homogeneity. Athey and Imbens (2018) propose an assumption that “restricts the heterogeneity of the treatment effects over time,” which implies $CATT_{e,\ell}$ can vary over e but not over ℓ . Borusyak and Jaravel (2017) refer to one type of treatment effects heterogeneity as “only across the time horizon,” which implies $CATT_{e,\ell}$ can vary over ℓ but not over e . Callaway and Sant’Anna (2020a) allow for “arbitrary treatment effect heterogeneity” when $CATT_{e,\ell}$ varies across cohorts and over time. Similarly, de Chaisemartin and D’Haultfœuille (2020) describe treatment effects that may be “heterogeneous across groups and over time periods.” Goodman-Bacon (2018) allow heterogeneous effects to either “vary across units but not over time” or “vary over time but not across units.” The literature has not converged on a single notion of treatment effects heterogeneity with time-varying treatment. Since researchers are interested in dynamic treatment effects when using a “dynamic” specification, we do not restrict the path of treatment effects but rather use “heterogeneity” to describe variation across cohorts only.

2.3 Relevance in the applied literature

To gauge the empirical relevance of our results, we survey the estimation methods used by the twelve papers collected by Roth (2019) from three leading economics journals that contain the phrase “event study” in their main text.¹ From this sample of applied papers, we learn what specifications empirical researchers are actually using when estimating two-way fixed effects regressions. Four papers in this sample consider the simple setting where units either receive their first treatment at the same time or never receive the treatment. The other eight papers in this sample consider the more complex setting where treated units receive their

¹We follow the selection criteria in Roth (2019): the original sample consists of 70 total papers, but is further constrained to these twelve papers with publicly available data and code. The data and code are used to determine exactly the specification estimated in these papers.